

ETHANOL — A SMALL MOLECULE WITH BIG INDUSTRIAL USES

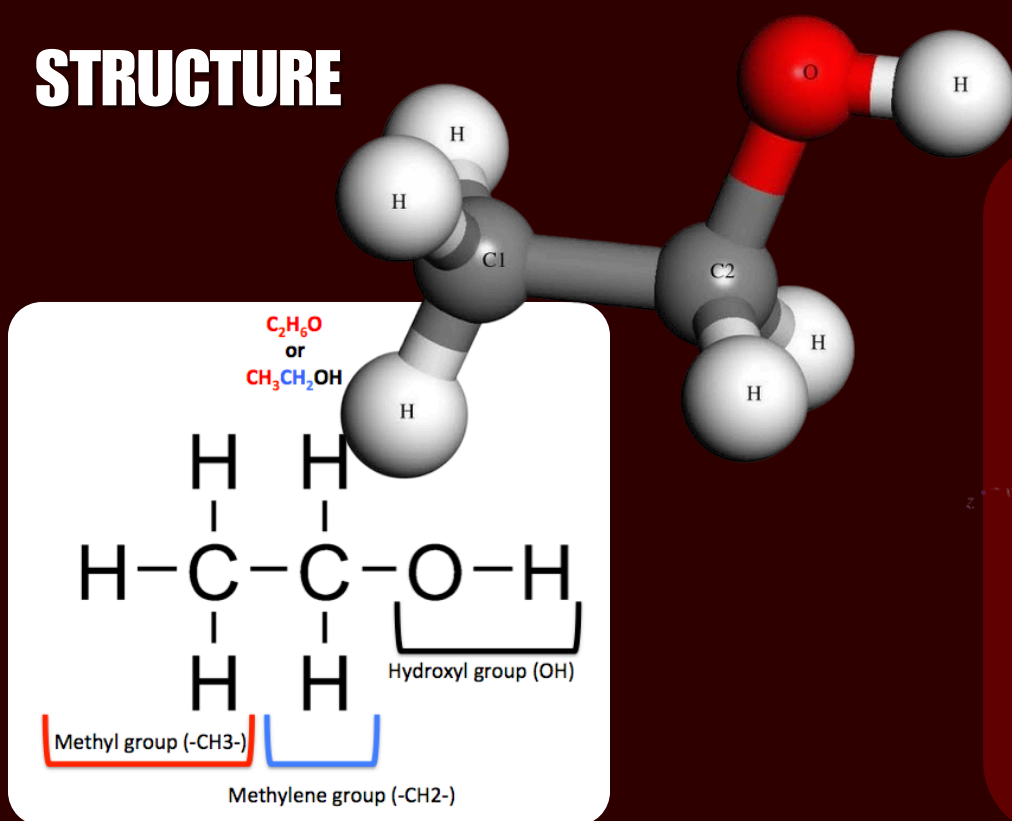
COMPOUND IDENTITY

Common name: Ethanol
IUPAC name: Ethanol
Molecular formula: C_2H_6O
Molar mass: 46.07 g/mol
CAS number: 64-17-5

IMPORTANT PROPERTIES

- Colorless liquid
- Polar Molecule
- Miscible with water
- Flammable
- Boiling point: $\sim 78.4\text{ }^{\circ}\text{C}$

STRUCTURE



USES & VALUE

- Solvent — used in chemical and pharmaceutical formulations
- Fuel component — used as a renewable fuel component
- Disinfectant formulations
- Chemical manufacturing
- Extraction and processing

WHAT IS AMAZING ABOUT IT?

Its small structure gives it a surprisingly wide range of uses.

Ethanol contains both a nonpolar ethyl group and a polar hydroxyl group. This combination helps it interact with both organic substances and water, making it useful as a solvent and in many formulations.

⚠ RISKS & SAFETY

⚠ Highly flammable liquid and vapor

- Keep away from ignition sources.
- Avoid excessive exposure to concentrated ethanol.
- Store and handle according to appropriate chemical-safety procedures.

CHEMICAL ENGINEERING CONNECTION

DISTILLATION & SEPARATION

ETHANOL+ WATER= SEPARATION BY DISTILLATION

Chemical engineers deal with ethanol-containing mixtures in distillation, separation, process design, and solvent recovery.

ONE SMALL ALCOHOL MOLECULE CONNECTS CHEMISTRY, SEPARATION PROCESSES, ENERGY, AND INDUSTRIAL PRODUCTION.

REFERENCES

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